

Quenzafil Film-Coated Tablets

Module 3 Quality Documentation — Formulation F-176/C

Reference Standards

Primary reference standards for the drug substance and specified impurities are qualified against compendial standards where available. Working standards are qualified against the primary standard under an approved protocol and are stored under controlled conditions.

Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Deviations observed during the campaign were investigated and closed with no impact to product quality.

Facilities and Equipment

Manufacturing is performed at the approved commercial site using dedicated product-contact equipment trains. Equipment qualification and cleaning validation are maintained under the site validation master plan. No changes to the approved facility registration are proposed.

Statistical evaluation of batch release data demonstrates process capability well within specification limits. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. Training records for personnel involved in manufacturing and testing are maintained under the site quality system.

Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. No changes to the approved analytical procedures are proposed in this submission. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable.

Stability Summary

Primary stability studies were conducted under ICH long-term (25°C/60% RH) and accelerated (40°C/75% RH) conditions on three registration batches. All quality attributes remained within acceptance criteria through the tested intervals. Photostability testing per ICH Q1B confirmed that the product is not photolabile in the commercial pack.

Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. The finished product is stored and distributed under

controlled ambient conditions consistent with the approved label storage statement. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection.

Process Validation Summary

Process performance qualification was executed on three consecutive commercial-scale batches. All predefined acceptance criteria for critical process parameters and critical quality attributes were met, supporting a conclusion of a validated state of control.

Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Deviations observed during the campaign were investigated and closed with no impact to product quality. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions.

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Cleaning verification swab results were below the health-based exposure limits derived for the product. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results.

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Section 3.2.P.4 — Excipients

For the commercial formulation F-176/C, Kosher Passover grade HyQual (Magnesium stearate, lubricant) is used.

The excipient control strategy was developed in accordance with Ph. Eur. general monograph 2034, USP-NF and FDA Guidance for Industry: Immediate Release Solid Oral Dosage Forms.

Commercial Control Strategy — Excipients

The current approved commercial specification (SPEC-EX-4074 Rev 06) lists Magnesium stearate as Ligamed MF-2-V.

Control of Drug Product

The drug product specification includes tests for description, identification, assay, uniformity of dosage units, dissolution, related substances, water content, and microbial limits. Analytical procedures are compendial where available; non-compendial procedures were validated in accordance with ICH Q2(R1).

Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. The control strategy links critical quality attributes to critical process parameters identified during development. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control.

Batch Formula

The representative commercial batch formula is provided for a nominal batch size of 400,000 tablets. Quantities of all components are expressed per tablet and per batch. Purified water is used as granulation fluid and is removed during processing; it does not appear in the final composition.

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